CBSE 2027 Maths — Predicted Paper (Set 4 of 10)

Based on the official blueprint and the highest-weightage topics identified in the latest CBSE Class 12 trends (such as the emphasis on calculus, 3D geometry, and competency-based questions), here is Set 4 of the predicted board paper.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let R be the relation in the set of real numbers $\mathbb{R}$ defined as $R = \{(a, b) : a \leq b\}$. Which of the following is true? (a) Reflexive and symmetric (b) Reflexive and transitive (c) Symmetric and transitive (d) Equivalence relation * **Chapter:** Relations and Functions * **Probability Tag: High**

Q2. Find the domain of the function $f(x) = \cos^{-1}(2x - 1)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Medium**

Q3. If A is a square matrix of order 3 such that $|A| = 8$, find the value of $|A \cdot adjA|$. * **Chapter:** Matrices and Determinants * **Probability Tag: Very High**

Q4. If A is a singular matrix of order 3, what is the value of the determinant of A? * **Chapter:** Determinants * **Probability Tag: High**

Q5. What is the derivative of a^x with respect to x (where $a > 0, a \neq 1$)? * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q6. The radius of a circle is r. What is the rate of change of the area of the circle with respect to its radius when $r = 6 \text{ cm}$? * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q7. Evaluate the indefinite integral: $\int (1 - x)\sqrt{x}\, dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q8. Find the area of the region bounded by the curve $y = x^2$, the x-axis, and the vertical line $x = 2$. * **Chapter:** Application of Integrals * **Probability Tag: Medium**

Q9. Write the degree of the differential equation: $x\left(\frac{d^2y}{dx^2}\right)^3 + y\left(\frac{dy}{dx}\right)^4 + x^3 = 0$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q10. Find a vector in the direction of the vector $\vec{a} = \hat{i} - 2\hat{j}$ that has a magnitude of 7 units. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q11. If $|\vec{a}| = 3$, $|\vec{b}| = 4$, and $|\vec{a} \times \vec{b}| = 6$, find the angle between the vectors $\vec{a}$ and $\vec{b}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q12. If a line has direction ratios $2, -1, -2$, determine its direction cosines. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q13. Find the z-coordinate of the point where the line passing through the points $A(3, 4, 1)$ and $B(5, 1, 6)$ crosses the XY-plane. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Medium**

Q14. The objective function $Z = 4x + 3y$ is to be maximized subject to certain constraints. The corner points of the feasible region are $(0, 0)$, $(40, 0)$, $(20, 30)$, and $(0, 50)$. Find the maximum value of Z. * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q15. If $P(A) = \frac{7}{13}$, $P(B) = \frac{9}{13}$, and $P(A \cap B) = \frac{4}{13}$, evaluate $P(A'|B)$. * **Chapter:** Probability * **Probability Tag: Very High**

Q16. Evaluate the definite integral: $\int_0^1 e^{2x}\, dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q17. Write the integrating factor (I.F.) for the linear differential equation: $\frac{dy}{dx} + y \sec x = \tan x$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q18. A die, whose faces are marked 1, 2, 3 in red and 4, 5, 6 in green, is tossed. Let A be the event "the number is even" and B be the event "the number is red". Are A and B independent? (Yes/No) * **Chapter:** Probability * **Probability Tag: Medium**

Q19. (Assertion-Reason): Assertion (A): If A and B are independent events, then $P(A|B) = P(A)$. **Reason (R):** For any two independent events, the probability of their intersection is the product of their individual probabilities, i.e., $P(A \cap B) = P(A)P(B)$. * **Chapter:** Probability * **Probability Tag: Very High**

Q20. (Assertion-Reason): Assertion (A): The tangent to the curve $y = x^3 - 3x$ at the point $x = 2$ has a slope of 9. **Reason (R):** The slope of the tangent to a curve $y = f(x)$ at any point x_1 is given by the derivative $f'(x_1)$. * **Chapter:** Application of Derivatives * **Probability Tag: High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Express $\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$ in the simplest form, where $0 < x < \pi$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q22. If $y = \sqrt{\sin x + y}$, find $\frac{dy}{dx}$. * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q23. Using vector methods, find the area of the triangle with vertices $A(1, 1, 1)$, $B(1, 2, 3)$, and $C(2, 3, 1)$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q24. Find the general solution of the differential equation: $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q25. Let A and B be independent events with $P(A) = 0.3$ and $P(B) = 0.4$. Find the probability that exactly one of the two events occurs. * **Chapter:** Probability * **Probability Tag: High**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^3 + x$ is a bijective function. * **Chapter:** Relations and Functions * **Probability Tag: Medium**

Q27. Express the matrix $A = \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix}$ as the sum of a symmetric and a skew-symmetric matrix. * **Chapter:** Matrices * **Probability Tag: Very High**

Q28. If $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$, find the value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}$. * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q29. Evaluate the integral using substitution and partial fractions: $\int \frac{2x}{(x^2+1)(x^2+3)}\, dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q30. Solve the linear differential equation: $x\frac{dy}{dx} + 2y = x^2 \log x$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q31. Bag I contains 4 red and 4 black balls, while Bag II contains 2 red and 6 black balls. One of the two bags is selected at random and a ball is drawn from the bag, which is found to be red. Find the probability that the ball was drawn from Bag I. * **Chapter:** Probability * **Probability Tag: Very High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Evaluate the definite integral using properties of integration: $\int_0^{\pi/4} \log(1 + \tan x)\, dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q33. Show that a closed right circular cylinder of a given total surface area and maximum volume is such that its height is equal to the diameter of its base. * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q34. Find the equation of the plane passing through the line of intersection of the planes $x + y + z = 1$ and $2x + 3y + 4z = 5$, and which is perpendicular to the plane $x - y + z = 0$. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: High**

Q35. Using integration, find the area of the region bounded by the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 6x$. * **Chapter:** Application of Integrals * **Probability Tag: High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (Factory Production) Two factories P and Q produce three types of boys' and girls' sports shoes. Factory P produces 50, 60, and 40 pairs of boys' shoes and 40, 50, and 60 pairs of girls' shoes in categories Type 1, Type 2, and Type 3 respectively. Factory Q produces 60, 40, and 50 pairs of boys' shoes and 50, 40, and 50 pairs of girls' shoes in the same respective categories. *(i)* Represent the production data of Factory P and Factory Q as 2×3 matrices X and Y. *(1 Mark)* *(ii)* Find the matrix representing the total combined production of each type of shoe by both factories. *(1 Mark)* *(iii)* If the profit per pair of boys' shoes is ₹100, ₹150, and ₹200 for Types 1, 2, and 3 respectively, and the profit for girls' shoes is ₹120, ₹160, and ₹220 respectively, use matrix multiplication to find the total profit generated by Factory Q. *(2 Marks)* * **Chapter:** Matrices * **Probability Tag: Medium**

Q37. Case Study 2: Linear Programming (Furniture Business) A furniture dealer deals in only two items: tables and chairs. He has ₹50,000 to invest and space to store at most 60 pieces. A table costs him ₹2,500 and a chair costs ₹500. He estimates that he can sell a table at a profit of ₹250 and a chair at a profit of ₹75. Let x be the number of tables and y be the number of chairs he purchases. *(i)* Formulate the mathematical constraints for his investment budget and storage capacity. *(1 Mark)* *(ii)* Write the objective function Z to maximize his profit. *(1 Mark)* *(iii)* Find the corner points of the feasible region and evaluate them to determine exactly how many tables and chairs he should buy to maximize his profit. *(2 Marks)* * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q38. Case Study 3: Probability (Total Probability & Bayes' Theorem) An insurance company believes that its clients can be divided into two classes: those who are accident-prone and those who are not. The company's historic statistics show that an accident-prone person will have an accident at some time within a fixed one-year period with a probability of 0.4. For a person who is not accident-prone, this probability is 0.2. The company knows that exactly 30% of the local population is accident-prone. *(i)* What is the total probability that a randomly chosen new policyholder will have an accident within a year of purchasing a policy? *(2 Marks)* *(ii)* If a new policyholder has an accident within a year of purchasing a policy, what is the exact probability that they are accident-prone? *(2 Marks)* * **Chapter:** Probability * **Probability Tag: Very High**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.